Geophys. J. R. astr. Soc. (1973) **32**, 53–63.

The Palaeomagnetism of the Foyers Plutonic Complex, Inverness-shire

Sylvia J. Kneen

(Received 1972 October 9)

Summary

A palaeomagnetic study of the Foyers Plutonic Complex, and its associated country rocks, has been made. From a total of 330 samples from 80 sites, 23 sites gave stable directions of magnetization after a/c cleaning. An α_{95} rejection criterion of 20° was applied to the directions of magnetization. The mean direction obtained was $D = 13{\cdot}6°$ $I = -5{\cdot}8°$ ($K = 4$, $\alpha_{95} = 17{\cdot}6$) giving a corresponding palaeomagnetic pole at 29° N 160° E and an oval of confidence $dP = 8{\cdot}9°$, $dm = 17{\cdot}6°$. Application of the conglomerate test to the overlying basal Devonian breccia, and a contact test to the surrounding schists, imply that the direction of magnetization obtained from the granite is original, and the pole position obtained is valid for Upper Silurian–Lower Devonian times.

1. Introduction

The work reported here forms part of a wider research project to examine the magnetic properties and sub-surface structure of certain Caledonian Newer granites. Interest in the Caledonian granites arose from an examination of the filtered aeromagnetic map of Britain (Hall & Dagley 1970) where it can be seen that certain granites are associated with strong positive aeromagnetic anomalies, whilst other granites have no magnetic expression at all. To investigate this further, gravity surveys, and sampling for magnetic work, have been completed in a number of 'granite' areas. In view of the scarcity of palaeomagnetic data for the British Silurian–Devonian, it was decided to investigate the palaeomagnetism of the granite complexes, in addition to other magnetic properties.

Palaeomagnetic studies of British Palaeozoic rocks have been reported by Chamalaun & Creer (1964), Creer (1966), Creer & Embleton (1967), Nesbitt (1967), Stubbs (1958), McMurry (1970) and Briden (1970). Storetvedt *et al.* (1966, 1968) and Lie *et al.* (1969) have reported results from Norwegian Palaeozoic formations.

2. Geology of the sampling area

The Geology of the Foyers Complex and surrounding schists has been described by Mould (1946), and more recently by Marston (1967, 1971). Only a brief summary is necessary here. The granite complex consists of three main members—quartz diorite, granodiorite, and adamellite, successively intruded into Moine Schists. (Fig. 1.) To the west, along Loch Ness side, basal Devonian breccia, and Devonian sandstone, unconformably overlie the 'granite' complex.

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

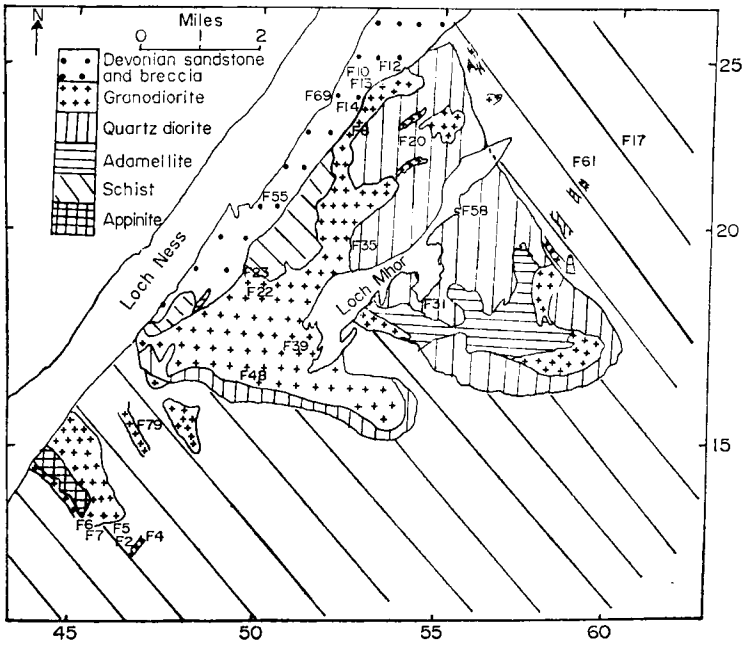


FIG. 1. Geological map of the sampling area, showing the location of the 23 accepted sampling sites. (Simplified from Mould 1946; Marston 1967).

Biotites from the complex have been dated by Miller & Brown (1965), using the potassium–argon method, and these yield an age of 400 ± 18 My. The radiometric evidence, and the structural relationship between the granite and the Moine schists, show that the Foyers Complex is post-tectonic with respect to the Caledonian orogeny, and is unlikely to have been affected by other tectonic events since its formation. Therefore the direction of magnetization in the granite should reflect the true direction of the geomagnetic field in Upper Silurian–Lower Devonian times.

3. Field procedure

A minimum of four cores were taken from each of 80 sites in the Foyers Plutonic Complex, and its associated country rocks. Cores, 5–8 cm long, were drilled using a portable two-stroke petrol motor, and oriented, *in situ*, relative to the Sun, or a known distant geographical feature. Eighty-six per cent of the cores were Sun-sighted.

4. Palaeomagnetic results

The oriented cores were sliced into 2·5-cm specimens for the purpose of measuring their remanent directions. The N.R.M. of each specimen was measured using a parastatic magnetometer. Stepwise a/c demagnetization was then carried out at 50-Oe intervals, up to a maximum of 1400 Oe; and the remanence of each specimen measured after each demagnetization step. The parastatic magnetometer is linked to a computer, which calculates the direction of magnetization immediately after each measurement, thus enabling the changes in direction to be plotted immediately. The demagnetization process was continued until a stable direction was attained.

Twenty-three sites gave stable directions of magnetization. These are listed in Table 1, together with the relevant statistics. The mean direction, for the separately

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

Table 1

Mean directions, pole positions, and Fisher statistics

Site code and rock type	*Dm*°	*Im*°	Number of samples (No. of specimens)	*N**	*R*	*K*	α_{95}°	North Pole (Latitude °N)	North Pole (Longitude °E)	*P*°	*M*°
F2 Contact schist	208·49	26·80	5(5)	10	9·9717	317	2·7	−14·82	326·92	1·60	2·95
F4 Monzonite	165·16	9·01	4(3)	6	5·8276	29	12·7	−27·10	12·17	6·44	12·77
F5 Biotite schist ..	174·40	27·46	4(5)	10	9·4640	17	12·1	−18·08	1·20	7·22	13·24
F6 Biotite schist ..	195·33	−18·31	4(4)	8	7·9286	98	5·6	−40·74	335·36	3·03	5·84
F7 Biotite schist ..	170·13	−1·36	4(6)	12	11·3043	16	11·3	−32·93	7·28	5·63	11·27
F8 Monzonite	177·51	3·39	4(3)	6	5·8285	29	12·6	−31·07	358·41	6·33	12·63
F12 Sandstone	239·63	−17·89	4(4)	8	7·8387	43	8·5	−23·85	286·86	4·57	8·82
F13 Sandstone	202·51	21·23	4(4)	8	7·7973	35	9·6	−19·33	332·03	5·30	10·07
F14 Sandstone	184·11	−3·03	4(3)	6	5·9075	54	9·2	−34·22	350·53	4·61	9·21
F17 Diorite	18·48	25·05	4(5)	10	9·6534	26	9·7	+43·76	150·20	5·58	10·38
F20 Quartz Diorite ..	205·43	19·83	4(5)	10	9·5259	19	11·4	−19·41	328·88	6·23	11·90
F22 Granodiorite	180·07	19·86	4(5)	10	9·8174	49	6·9	−22·56	355·43	3·80	7·27
F23 Granodiorite	200·08	71·17	4(4)	8	7·2318	9	19·4	+24·07	343·27	29·55	33·85
F31 Granodiorite	197·64	71·07	4(4)	8	7·9253	94	5·8	+23·62	344·71	8·74	10·03
F35 Monzonite	195·23	19·29	4(4)	8	7·4350	12	16·4	−21·72	339·33	8·92	17·10
F39 Biotite schist ..	32·33	−13·94	4(4)	8	7·7195	25	11·3	+20·53	140·98	5·91	11·56
F48 Granodiorite	233·99	18·11	4(4)	8	7·7748	31	10·1	−10·29	301·28	5·44	10·48
F55 Sandstone	182·24	−11·05	4(4)	8	7·9514	144	4·6	−38·35	352·66	2·38	4·69
F58 Granodiorite	151·68	40·62	4(3)	6	5·8236	28	12·8	−6·14	21·51	9·38	15·50
F61 Granite	145·30	44·71	4(4)	8	7·8974	68	6·8	−1·51	26·19	5·37	8·52
F69 Granite	221·97	2·80	4(4)	8	7·7720	31	10·2	−22·47	309·16	5·09	10·17
F70 Sandstone	219·28	8·97	4(3)	6	5·9214	64	8·5	−20·61	313·11	4·31	8·54
F79 Granodiorite	186·58	21·39	4(4)	8	7·6883	22	11·9	−21·50	348·55	6·63	12·59

Co-ordinates of sampling site: 57·18 °N355·5 °E.

*N = Twice the number of specimens measured

Mean direction and pole position from all sites

D°	I°	K	α_{95}°	Pole Latitude °N	Pole Longitude °E	*dP*°	*dM*°
3·6	−5·8		17·58	29	160	8·9	17·6

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

oriented samples from each site was computed by the method of Dagley & Ade-Hall (1970). An α_{95} rejection criterion of 20° was applied to the data.

5. Evidence for the ' magnetic age ' of the samples

5.1 *The conglomerate test*

The directions of magnetization of specimens from the basal Devonian breccia were used to apply the conglomerate test to the magnetization of the granite complex (Graham 1949). Twenty-eight samples of basal Devonian breccia were collected from seven sites. Along the line of the Great Glen Fault, the breccia consists of very small (approx. 1 cm^2 cross-section) fragments in a sandstone matrix. The haematite in the sandstone is the main carrier of remanence, and the directions of magnetization of these specimens all lie in the same quadrant, but do exhibit a small, significant scatter (Fig. 2). Specimens from the same sample also show different demagnetization

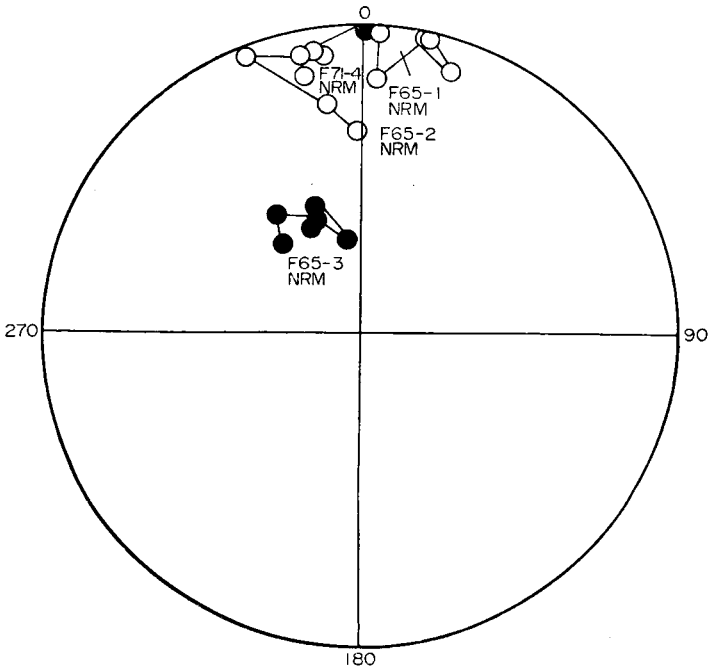


FIG. 2. Demagnetization characteristics of fine-grained breccia from sites F65, F71, F72. ○ positive inclination; ● negative inclination.

characteristics, and different directions of magnetization. Further to the east, larger fragments can be found in the breccia; and it was possible to obtain specimens from granite pebbles in the breccia. The directions of magnetization obtained from these specimens show a large degree of scatter (Fig. 3).

The Devonian sandstone, which overlies the breccia, gave stable directions, on demagnetization, and these directions are consistent with those obtained from the granite.

These results indicate that the direction of magnetization of the granitic rocks is a primary direction.

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

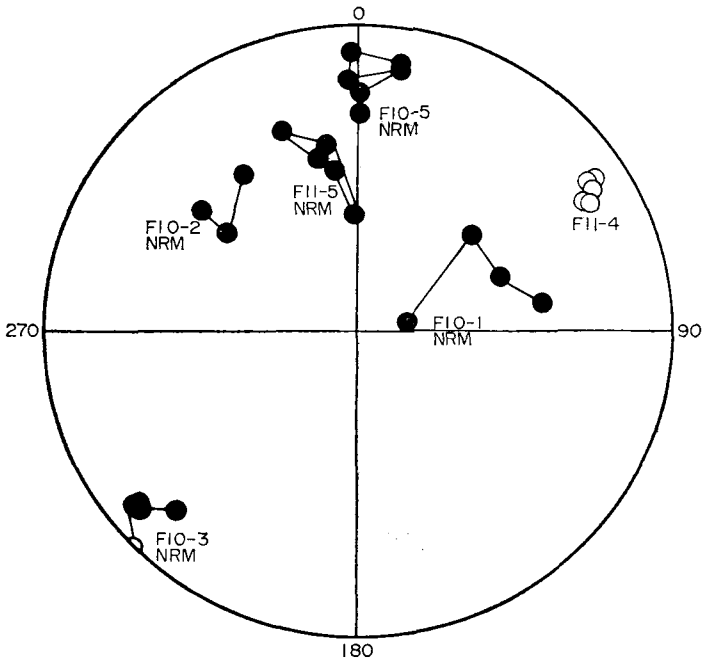


FIG. 3. Demagnetization characteristics of course-grained breccia, and granite pebbles from the breccia (Sites F10 and F11). ○ positive inclination; ● negative inclination; 10–1 Microgranite pebble from Breccia; 10–2 Contains a large granite pebble plus smaller pieces; 10–3 Fairly fine grained, containing fragments of granite and schist; 10–5 Granite pebble from Breccia; 11–4 Contains granitic fragments, and a large piece of biotite schist; 11–5 Contains small fragments.

5.2 *A contact test*

Specimens from the Schistose country rocks near to the granite boundary yield a direction of magnetization in close agreement to the stable direction obtained from the granite complex. The contact schists are magnetically hard, and show a high stability in fields up to 1400 Oe (Fig. 4(a) and (b)). These results also indicate that the direction of magnetization in the granite is original.

6. Opaque petrology

Polished sections from the granite complex, the Moine schists, and the Devonian sandstones, show that the most important magnetic mineral is haematite.

6.1 *Moine schists from near the 'granite' contact*

Typically these contain small ($\leqslant 10\mu$) haematite crystals, containing lamellae of ilmenite. Sharp, straight, rutile needles, which taper to a point, are seen crossing the ilmenite lamellae. The presence of the small haematite crystals accounts for the observed high stability against a/c demagnetization.

6.2 *The members of the granite complex*

These sections reveal a large number of single, small, haematite crystals; and a small number of very large titanomagnetite crystals ($\geqslant 100\mu$), containing varying amounts of unexsolved titanium. The demagnetization characteristics of these rocks

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

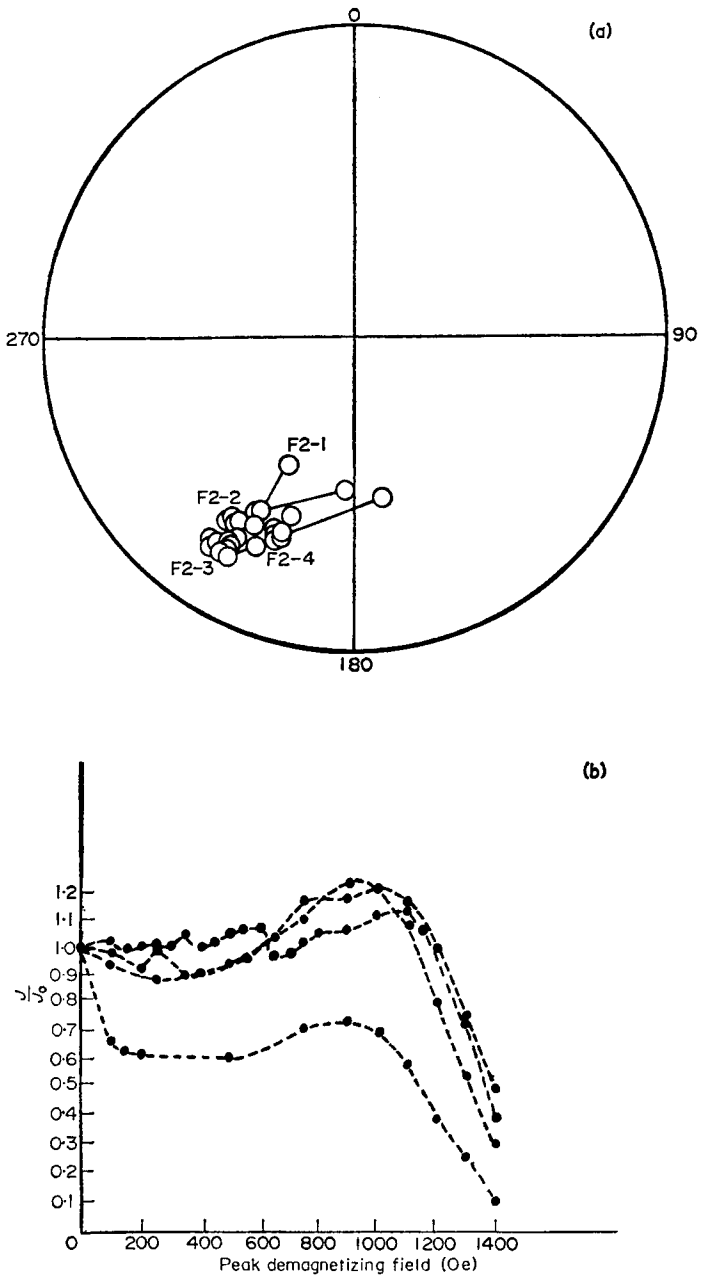


FIG. 4(a). Demagnetization characteristics of contact schist F2, in fields of 0–1400 Oe. ○ positive inclination. (b) Normalized a/c demagnetization curves for contact schist F2.

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

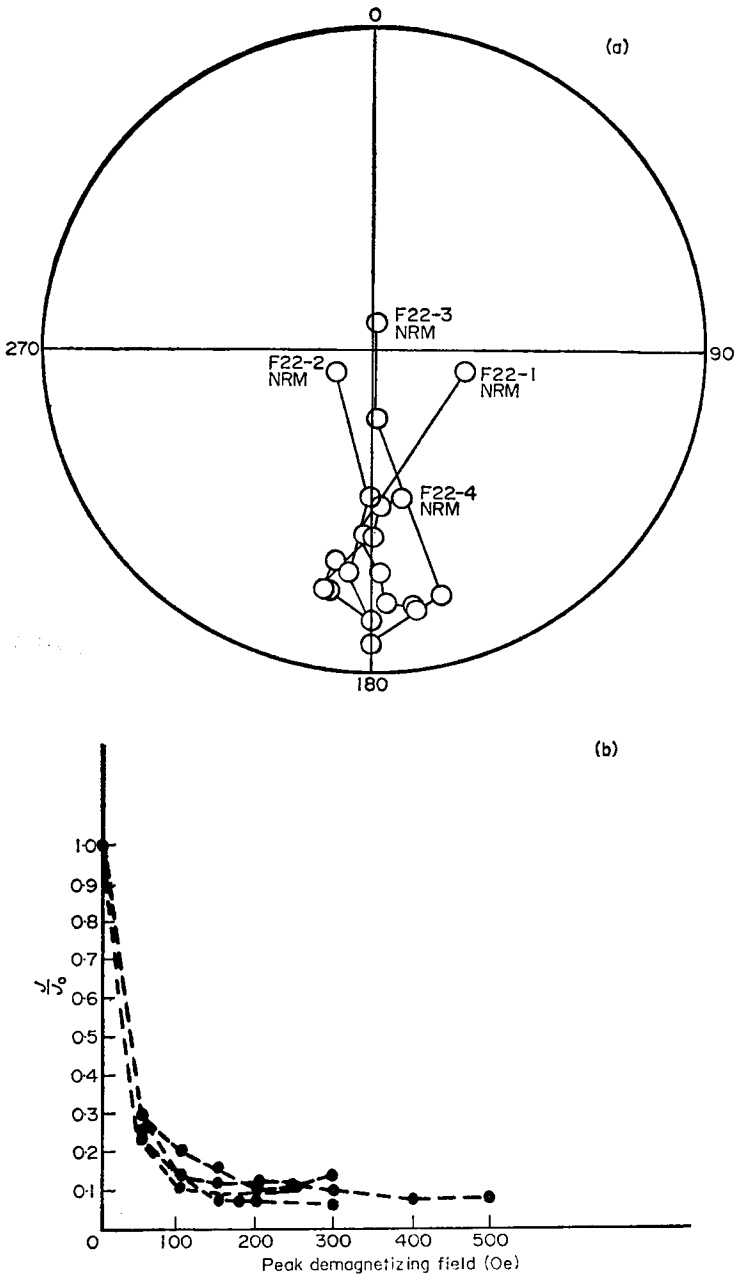


FIG. 5(a). Demagnetization characteristics for site F22. ○ positive inclination; ● negative inclination. (b) Normalized a/c demagnetization curves for site F20.

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

reflect their petrology. The specimens show a large shallowing of inclination, and a 90 per cent reduction in intensity on a/c demagnetization in fields up to 100 Oe. Demagnetization in higher fields produces only small changes; and a stable direction is achieved (Fig. 5(a) and (b)).

The initial change in direction is attributed to a component directed along the present magnetic field. This soft component must lie in the large, multi-domain titanomagnetite crystals. The much weaker, stable component, resides in the small haematite crystals.

When samples containing large titanomagnetite grains which had already been demagnetized, were stored overnight, it was found that they acquired a strong magnetization, which was easily removed by low demagnetizing fields. This also demonstrates that the greater part of the magnetization of the titanomagnetite is extremely soft.

6.3 *Devonian sandstones*

The Devonian sandstones contain numerous small, individual haematite grains, having good crystal form; and also collections of anhedral haematite grains. These two generations of haematite appear to be primary haematite, and secondary haematite formed as a result of silicate alteration. The time interval between the two generations of haematite is probably not magnetically significant; and both should reflect an early Devonian direction of magnetization.

7. Discussion of results

The pole positions obtained from the 23 accepted sites are shown in Fig. 6. For these sites the mean direction is $D = 13{\cdot}6°$ $I = -5{\cdot}8°$ ($K = 4$, $\alpha_{95} = 17{\cdot}6$) giving a corresponding mean pole at 29° N 160° E (Northern Hemisphere co-ordinates) and an oval of confidence $dP = 8{\cdot}9°$ $dm = 17{\cdot}6°$. The different rock types are given different

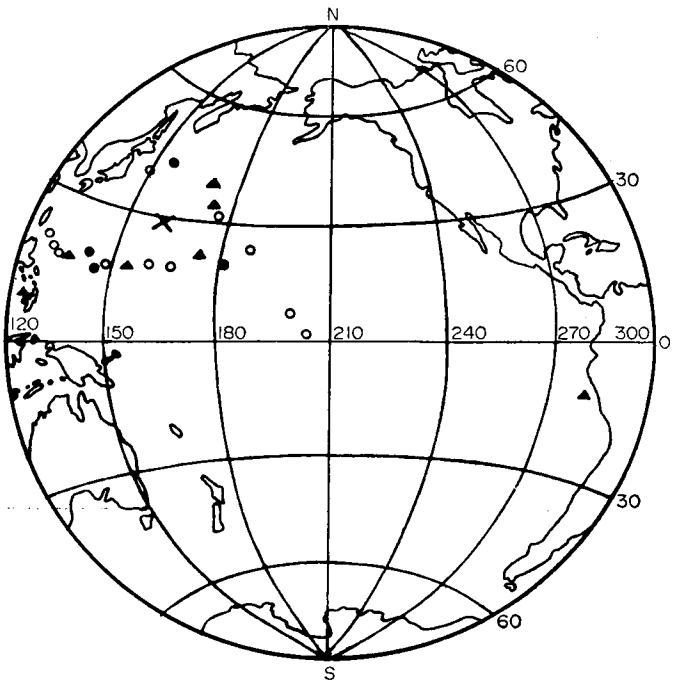


FIG. 6. Pole positions obtained from the 23 accepted sites. ● schist; ○ granite complex; ▲ sandstone; × mean pole.

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

symbols in Fig. 6, and it can be seen that there is no obvious co-relation between rock type and pole position. Electron microprobe studies of the ulvospinel content of the titanomagnetites, to obtain the temperature of crystallization, and oxygen fugacity; coupled with more extensive sampling, may help to elucidate a time sequence between pole positions.

Using the tables of pole positions compiled by McElhinny (1967, 1968, 1969, 1970, 1972), the Cambrian, Ordovician, Silurian, and Devonian mean pole positions, relative to Britain, and to Scandinavia, have been plotted in Fig. 7. The mean pole

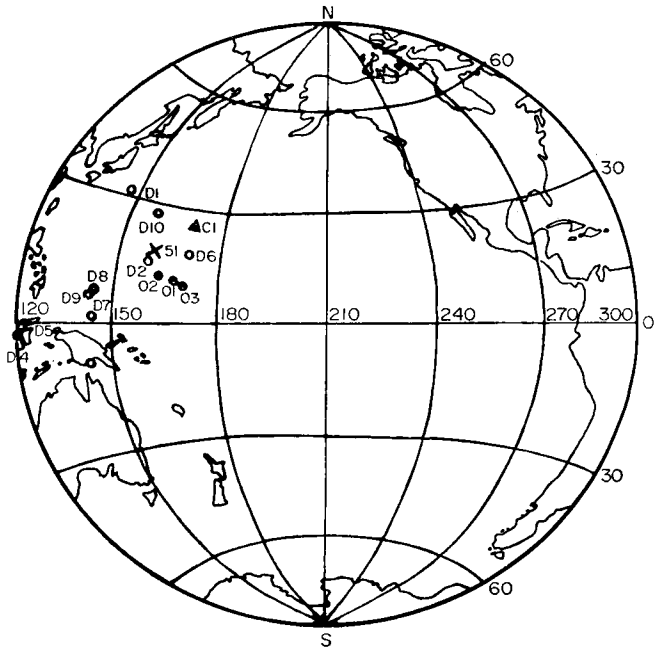


FIG. 7. British and Norwegian Palaeozoic pole positions. ▲ Cambrian; ○ Ordovician; × Silurian; ● Devonian. C1 Briden *et al*, (1970) 01, 02 and 03 Nesbitt (1967); S1 Storetvedt et al. (1968); D1 Creer (1966); D2 Storetvedt & Gjellestad (1966); D3 Stubbs (1958); D4 Chamalaun & Creer (1964); D5 Creer & Embleton (1967); D6 Lie *et al.* (1969); D7 McMurry (197); D8 and D9 Briden (1970); D10 Foyers from present study.

from the present study is also included (Pole D10), and it can be seen that this pole shows closer agreement to the Norwegian poles (S1, D2 and D6) than to the British data. A possible interpretation of the Devonian data would be that the North of Scotland and Norway formed part of the same tectonic plate, which was separated from the European plate. This is thought to be unlikely however, as Briden's Cambrian pole (from the Caerfai series, Wales) and Nesbitt's Ordovician Poles (from the Borrowdale volcanics) are in the same locality as the Devonian poles of Storetvedt and the present study. Another interpretation would be that the palaeomagnetic pole relative to Britain and Norway remained in approximately the same locality throughout Cambrian Ordovician Silurian and Devonian times, apart from a short rapid drift episode in Upper Silurian–Lower Devonian times represented by poles D3, D4, D5, D7, D8, and D9. A possible polar wander path would go from pole C1 (Cambrian) to Nesbitt's Ordovician poles, followed by a rapid shift to the vicinity of pole D7 and end near Creer's pole (D1) in the Upper Devonian–Lower Carboniferous. This is similar to the hypothesis of Briden (1967) for the movement of Gondwanaland.

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

Interpretations based on widespread re-magnetization of British Palaeozoic rocks have been advanced by Creer (1968). The evidence from Foyers indicates that remagnetization has not occurred here, but a comprehensive survey of the whole of the British Palaeozoic would be necessary before this could be established with certainty.

Acknowledgments

I would like to thank Professor R. L. Wilson, Dr P. Dagley and Dr G. C. Brown for many useful discussions, advice on the opaque petrology, and a critical appraisal of the manuscript. I also thank Dr R. J. Marston and Sheffield University for permission to use unpublished maps from Dr Marston's thesis. The work is done under a N.E.R.C. research grant, and whilst in receipt of a N.E.R.C. studentship.

Sub-Department of Geophysics,
Oliver Lodge Laboratory,
University of Liverpool.

References

Briden, J. C., 1967. Recurrent Continental Drift of Gondwanaland, *Nature Lond.*, **215,** 1334–1339.

Briden, J. C., 1970. Palaeomagnetic results from the Arrochar, and Garabel Hill–Glen Fyne igneous complexes, Scotland. *Geophys. J. R. astr. Soc.*, **21,** 457–469.

Briden, J. C., Irons, J. & Johnson, P. A., 1970. Palaeomagnetic studies of the Caerfai Series and the Skomer Volcanic Group (Lower Palaeozoic, Wales), *Geophys. J. R. astr. Soc.*, **22,** 1–16.

Chamalaun, F. H. & Creer, K. M., 1964. Thermal demagnetization studies on the Old Red Sandstone of the Anglo-Welsh Cuvette. *J. geophys. Res.*, **69,** 1607–1616.

Creer, K. M., 1966. Palaeomagnetic studies of basic sills and dykes from South-West England, *Geophys. J. R. astr. Soc.*, **11,** 415–422.

Creer, K. M., 1968. Palaeozoic Palaeomagnetism, *Nature Lond.*, **219,** 246–250.

Creer, K. M. & Embleton, B. J. J., 1967. Devonian Palaeomagnetic pole for Europe and North America, *Nature Lond.*, **214,** 42–43.

Dagley, P. & Ade-Hall, J. M., 1970. Cretaceous, Tertiary, and Quaternary Palaeomagnetic results from Hungary, *Geophys. J. R. astr. Soc.*, **20,** 65–87.

Graham, J. W., 1949. The stability and significance of magnetism in sedimentary rocks. *J. geophys. Res.*, **54,** 131–167.

Hall, D. & Dagley, P., 1970. An analysis of the smoothed aeromagnetic map of Great Britain and Northern Ireland, *N.E.R.C. I.G.S., Rep. No.* 70/10. *H.M.S.O.*

Lie, L. G., Storetvedt, K. M. & Gjellestad, G., 1969. Palaeomagnetism of the Kvamshesten Old Red sequence, South-west Norway, *Nord. Geol. Tidsskr.*, **49,** 241–249.

McElhinny, M. W., 1967. Palaeomagnetic directions and pole positions VIII, *Geophys. J. R. astr. Soc.*, **15,** 409–430.

McElhinny, M. W., 1968. Palaeomagnetic directions and pole positions IX, *Geophys. J. R. astr. Soc.*, **16,** 207–224.

McElhinny, M. W., 1969. Palaeomagnetic directions and pole positions X, *Geophys. J. R. astr. Soc.*, **18,** 305–327.

McElhinny, M. W., 1970. Palaeomagnetic directions and pole positions XI, *Geophys. J. R. astr. Soc.*, **20,** 417–429.

McElhinny, M. W., 1972. Palaeomagnetic directions and pole positions XII, *Geophys. J. R. astr. Soc.*, **27,** 237–257.

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026

McMurry, E., 1970. Palaeomagnetic results from Scottish lavas of Lower Devonian Age, *Palaeogeophysics*, 253–262, ed. Runcorn S. K. Academic Press, London.

Marston, R. J., 1967. Unpublished Ph.D. Thesis, University of Sheffield.

Marston, R. J., 1971. The Foyers Granitic Complex, Inverness-shire, Scotland. *Q. Jl geol. Soc.*, **126,** 331–368.

Miller, J. A. & Brown, P. E., 1965. Potassium–Argon studies in Scotland, *Geol. Mag.*, **102,** 106–134.

Mould, D. D. C. P., 1946. The Geology of the Foyers Granite and the surrounding country, *Geol. Mag.*, **83,** 249–265.

Nesbitt, J. D., 1967. Palaeomagnetic evidence for the Ordovician geomagnetic pole position, *Nature Lond.*, **216,** 49–50.

Storetvedt, K. M. & Gjellestad, G., 1966. Palaeomagnetic investigation of an Old Red Sandstone formation of Southern Norway, *Nature Lond.*, **212,** 59–61.

Storetvedt, K. M., Halvorsen, E. & Gjellestad, G., 1968. Thermal Analysis of the N.R.M. of some Upper Silurian sandstones of the Oslo region, *Tectonophys.*, **5,** 413–426.

Stubbs, P. H. S., 1958. Ph.D. Thesis, University of London.

Downloaded from https://academic.oup.com/gji/article/32/1/53/617795 by Curtin University Library user on 26 May 2026